Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

# Collecting Like Terms

Name _____ Date _____ No calculator Version 1

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**KEY WORDS** term *(tūnga)* · like terms *(tūnga ōrite)* · coefficient *(tau whakarea)* · variable *(taurangi)*

## A One letter ●○○

**Remember:** Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$.

**1.** Simplify.

**a)** $3x + 5x$ = _____ **b)** $7a - 2a$ = _____ **c)** $4y + y$ = _____ **d)** $9m - 9m$ = _____

**e)** $2x + 3x + 4x$ = _____ **f)** $8p - 3p + p$ = _____ **g)** $5k - 8k$ = _____ **h)** $-2n + 7n$ = _____

**i)** $6b + 6b - 5b$ = _____ **j)** $-3t - 4t$ = _____ **k)** $10w - w - 6w$ = _____ **l)** $c + c + c$ = _____

**m)** $2d + 9d$ = _____ **n)** $12z - 7z$ = _____ **o)** $3h + 3h + 3h$ = _____ **p)** $-5r + 5r$ = _____

**q)** $15x - 6x - 4x$ = _____ **r)** $q + 8q$ = _____ **s)** $-v - v$ = _____ **t)** $20e - 11e$ = _____

## B Two letters ●●○

**Watch out:** Terms with different letters cannot be added: $3x + 2y$ stays as it is.

**2.** Simplify by collecting like terms.

**a)** $3x + 2y + 4x$ = _____ **b)** $5a + 3b - 2a$ = _____ **c)** $6m + 4n + 2n$ = _____

**d)** $7p - 3q - 4p$ = _____ **e)** $2x + 5y + 3x + y$ = _____ **f)** $8a + 2b - 3a - 5b$ = _____

**g)** $4c - 6d + 2c + 9d$ = _____ **h)** $x + y + x + y$ = _____ **i)** $9h - 2k - 9h + 5k$ = _____

**j)** $-3a + 4b + 7a - 6b$ = _____ **k)** $5x - 2y - 8x + 3y$ = _____ **l)** $6u + 6v - 6u - v$ = _____

**m)** $10x - 3y - 4x - 2y$ = _____ **n)** $2s + 7t + 5s - 7t$ = _____ **o)** $a - b + a - b$ = _____

**p)** $12m + 5n - 15m + n$ = _____ **q)** $3x + 3y + 3x - 3y$ = _____ **r)** $-4g - 2h + 9g + 6h$ = _____


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## C Terms with powers and numbers ●●○

⚠ **Watch out:** $x^2$ and $x$ are not like terms. A number on its own can only be added to another number.

**3.** Simplify.

**a)** $3x^2 + 2x^2$

**b)** $5x + 4 + 2x + 3$

**c)** $4x^2 + 3x + 2x^2$

**d)** $7a + 9 - 2a - 5$

**e)** $6y^2 - 2y - 4y^2 + 5y$

**f)** $2x^2 + 3x + 1 + 4x^2 - x$

**g)** $10 + 3m - 4 + 2m$

**h)** $a^2 + a + a^2 + a$

**i)** $5p^2 - 5p - 2p^2 - 3p$

**j)** $8x^2 - 3x^2 + x^2$

**k)** $2t^2 + 6t + 5 - 2t^2$

**l)** $9 + 4k - 9 - 4k$

**m)** $3n^2 + 7 + 2n - n^2 - 10$

**n)** $x^3 + 2x^2 + 3x^3$

**o)** $5w - 2w^2 + 3w + 6w^2$

## D Mixed practice ●●●

**4.** Simplify.

**a)** $3ab + 2ab$

**b)** $4xy + 3x + 2xy$

**c)** $5a + 2b + 3c + a - 2c$

**d)** $7x - 3y + 4 - 2x + 5y - 1$

**e)** $2mn + 5m - mn - 5m$

**f)** $6a^2 + 2ab - 3a^2 + 4ab$

**g)** $0.5x + 1.5x + 2y$

**h)** $9p - 4q - 9p + 4q$

**i)** $3x^2 - x + 8 - 3x^2 + 6x - 8$

**j)** $2xy + 3yz + 4xy - yz$

**k)** $5a - 2ab + 3b + 2ab$

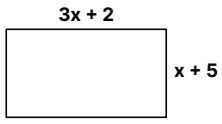
**l)** $x^2y + 4xy + 2x^2y$

## E Perimeter ●●●

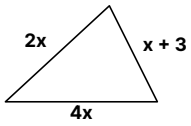
💡 **Remember:** The perimeter is the distance around a shape. Add all the sides, then collect like terms.

**5.** Write an expression for the perimeter of each shape. Simplify your answer.

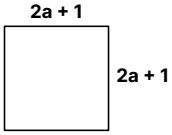
**a)**



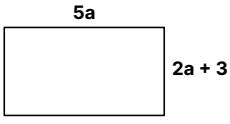

**b)**



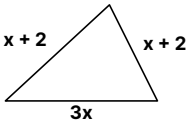

**c)**



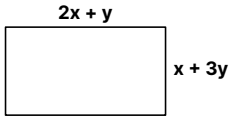

**d)**




**e)**




**f)**




## ★ Extension

**6.** Simplify $3x + 2y - (1x - 4y)$. Take care with the signs inside the bracket.

**7.** A rectangle has a perimeter of $10x + 6$. Its length is $3x + 1$. Find an expression for its width.



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# Collecting Like Terms

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**KEY WORDS** term *(tūnga)* · like terms *(tūnga ōrite)* · coefficient *(tau whakarea)* · variable *(taurangi)*

## A One letter ●○○

**Remember:** Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$.

**1.** Simplify.

**a)** $4x + 2x =$ _____ **b)** $9a - 4a =$ _____ **c)** $6y + y =$ _____ **d)** $5m - 5m =$ _____

**e)** $3x + 2x + 5x =$ _____ **f)** $7p - 2p + p =$ _____ **g)** $4k - 9k =$ _____ **h)** $-3n + 8n =$ _____

**i)** $5b + 5b - 4b =$ _____ **j)** $-2t - 6t =$ _____ **k)** $12w - w - 5w =$ _____ **l)** $d + d + d =$ _____

**m)** $3c + 8c =$ _____ **n)** $14z - 6z =$ _____ **o)** $4h + 4h + 4h =$ _____ **p)** $-7r + 7r =$ _____

**q)** $16x - 5x - 7x =$ _____ **r)** $q + 9q =$ _____ **s)** $-u - u =$ _____ **t)** $25e - 13e =$ _____

## B Two letters ●●○

**Watch out:** Terms with different letters cannot be added: $3x + 2y$ stays as it is.

**2.** Simplify by collecting like terms.

**a)** $4x + 3y + 2x =$ _____ **b)** $6a + 2b - 3a =$ _____ **c)** $5m + 3n + 4n =$ _____

**d)** $8p - 2q - 5p =$ _____ **e)** $3x + 4y + 2x + y =$ _____ **f)** $9a + 3b - 4a - 7b =$ _____

**g)** $5c - 8d + 3c + 10d =$ _____ **h)** $a + b + a + b =$ _____ **i)** $7h - 3k - 7h + 6k =$ _____

**j)** $-2a + 5b + 9a - 8b =$ _____ **k)** $4x - 3y - 9x + 5y =$ _____ **l)** $8u + 8v - 8u - v =$ _____

**m)** $11x - 2y - 5x - 3y =$ _____ **n)** $3s + 8t + 4s - 8t =$ _____ **o)** $c - d + c - d =$ _____

**p)** $10m + 4n - 14m + n =$ _____ **q)** $5x + 5y + 5x - 5y =$ _____ **r)** $-3g - 4h + 8g + 9h =$ _____


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## C Terms with powers and numbers ●●○

⚠ **Watch out:** $x^2$ and $x$ are not like terms. A number on its own can only be added to another number.

**3.** Simplify.

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**b)** $6x + 2 + 3x + 5$

**c)** $5x^2 + 2x + 3x^2$

**d)** $8a + 7 - 3a - 4$

**e)** $7y^2 - 3y - 5y^2 + 6y$

**f)** $3x^2 + 4x + 2 + 2x^2 - x$

**g)** $12 + 4m - 5 + 3m$

**h)** $b^2 + b + b^2 + b$

**i)** $6p^2 - 4p - 3p^2 - 2p$

**j)** $9x^2 - 4x^2 + x^2$

**k)** $3t^2 + 5t + 4 - 3t^2$

**l)** $7 + 6k - 7 - 6k$

**m)** $4n^2 + 5 + 3n - n^2 - 9$

**n)** $x^3 + 4x^2 + 2x^3$

**o)** $6w - 3w^2 + 2w + 5w^2$

## D Mixed practice ●●●

**4.** Simplify.

**a)** $4ab + 3ab$

**b)** $5xy + 2x + 3xy$

**c)** $6a + 3b + 2c + a - c$

**d)** $8x - 2y + 5 - 3x + 6y - 2$

**e)** $3mn + 4m - 2mn - 4m$

**f)** $7a^2 + 3ab - 2a^2 + 5ab$

**g)** $1.5x + 2.5x + 3y$

**h)** $7p - 5q - 7p + 5q$

**i)** $4x^2 - x + 6 - 4x^2 + 7x - 6$

**j)** $3xy + 2yz + 5xy - yz$

**k)** $6a - 3ab + 2b + 3ab$

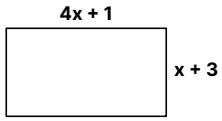
**l)** $x^2y + 5xy + 3x^2y$

## E Perimeter ●●●

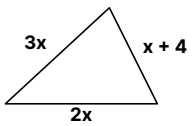
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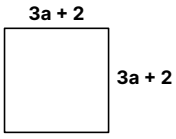
**a)**



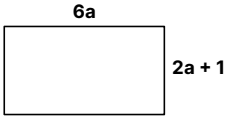

**b)**



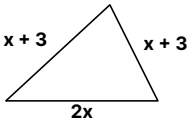

**c)**



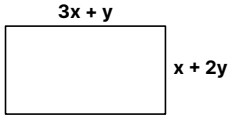

**d)**




**e)**




**f)**




## ★ Extension

**6.** Simplify $5x + 3y - (2x - 6y)$. Take care with the signs inside the bracket.

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